PARTH TAWDE

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SUMMARY

Python Software Developer and MSc Data Science graduate with experience in Python application development, Django, React, computer vision, image processing and design automation. Currently developing product personalisation solutions using Python, OpenCV, Pillow, Adobe tools and Playwright. Experienced in AI-powered applications, machine learning systems and automation workflows.

SKILLS

Programming: Python, SQL, PySpark
Web Dev: Django, React, FastAPI, Node.js, Express, TS
Computer Vision: OpenCV, Pillow, OCR, Image Processing
Machine Learning: TensorFlow, PyTorch, Keras, Scikit-learn
Generative AI: LangChain, LangGraph, Ollama, GLINER, Prompt Eng.
Automation: Playwright, Adobe, Product Personalisation, Design Auto.
Data: Pandas, NumPy, Matplotlib, Seaborn, Power BI
Tools: Git, MongoDB, Jupyter

EXPERIENCE

Nijikart — **Mumbai, India**
Software Python Developer — *01/2026 – Current*

- Build application features with Django and React, integrating backend processing with interactive workflows.
- Develop Python software and product personalisation workflows for digital products.
- Create personalised product assets using OpenCV, Pillow, Python libraries and Adobe tools.
- Use Playwright to automate design operations on products, turning repetitive manual work into scalable browser-based workflows.

NeuIQ — **Mumbai, India**
Analyst — *10/2025 – 01/2026*

- Benchmarked model performance using statistical correlation metrics and evaluated proposed solutions.
- Produced LLM-generated open-ended responses and reports/dashboards to assess real vs. synthetic data quality.
- Generated CTGAN-based synthetic structured data aligned with real schemas while preserving distributions and relationships.

Pinnotech Systems LLP — **Thane, India**
Data Science Intern — *01/2023 – 08/2023*

- Built a VGG19 transfer-learning model for plant disease classification using preprocessing, augmentation and feature extraction.
- Developed YOLO-based helmet, head and face detection using custom annotated datasets and mAP/IoU evaluation.

PROJECTS

AI-Powered Document Management System

- Built a full-stack AI-powered Document Management System using React, TypeScript, Node.js, Express and MongoDB for document ingestion, organisation and retrieval.
- Developed OCR workflows for handwritten documents, scanned PDFs and images using Python OCR technologies and structured processing pipelines.
- Implemented local AI using Qwen 2.5 via Ollama and GLINER to generate summaries, extract named entities, create tags, classify documents and recommend filenames without relying on external APIs.
- Designed metadata extraction and indexing pipelines for AI-generated summaries, entities, tags and document attributes, enabling intelligent search and filtering.
- Designed independent Python/FastAPI microservices for OCR, NER and table extraction to improve fault isolation and maintainability.

Skin Lesion Classification

- Designed a CNN in TensorFlow and fine-tuned a Vision Transformer in PyTorch for melanoma classification
- Applied oversampling and weighted loss functions to address class imbalance.

High-Resolution Satellite Imagery Analysis

- Developed CNN/U-Net models for land-cover, vegetation and urban-feature extraction.
- Generated DEM, DTM and DSM for terrain reconstruction.
- Applied super-resolution and patch-wise training for large imagery.

Netflix Recommendation System

- Built collaborative filtering using Probabilistic Matrix Factorization with SGD, L2 regularisation and Bayesian priors.
- Tuned latent dimensions to improve RMSE.

EDUCATION

University of Hertfordshire **United Kingdom**
Master of Science (M.Sc.) in Data Science — First Class Distinction *2024 – 2025*

MIT ADT University **Pune, India**
Bachelor of Technology (B.Tech) in Computer Science Engineering — CGPA: 6.94/10 *2019 – 2023*

PUBLICATION & ACTIVITIES

Publication: High-Resolution Satellite Imagery Analysis for Terrain and Surface Data Extraction: Techniques and Applications, 2023.

Medical Imaging Data Science Club (Clustered Minds) *(Feb – Jun 2025):* Designed retinal image enhancement using K-Means + Fuzzy C-Means with spatial features. Earned Judges' Choice 4th place in a university-level data science competition.